

SCIENTIFIC METHODOLOGY AND (EMPIRICAL|EXPERIMENTAL) EVALUATION

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Master 2 MOSIG
2026-2027



OVERVIEW

TEACHERS AND CONTACT



Resources

- Github (schedule, slides, homeworks)
- MOOC (Learn reproducible research by yourself)

Communication

- Pad (notes and homework reporting)
- Mattermost (questions, references, ...)

TEN REASONS WHY GENAI IS BANNED FROM THIS LECTURE

Why you shouldn't use GenAI for your studies

1. **Training and Cognitive Deficit** You're here to learn, not to produce. AI may accelerate production but it undermines training: by letting AI think and produce for you, you're not exercising your own abilities
Studies on the subject are starting to become available
2. **Occupation** Jobs that involve verifying content produced by AI are demeaning, exhausting, stressful, and depressing. Getting into the habit of working with AI during your studies means running the risk of no longer feeling capable of doing anything else and spending your professional life in a way that is not fulfilling
3. **Satisfaction** Succeeding on your own, without AI, brings pleasure and a sense of confidence in your own abilities
4. **Reciprocity** I will never use AI to create content for you, nor will I ever correct your work using AI. Reading AI-generated content or realizing that you've used it saddens me and undermines my trust

Why you shouldn't use GenAI at all

5. **Working Conditions** The growth of GenAI relies on the exploitation of hundreds of millions of clickworkers
6. **Intellectual Property** AI is based on the illegal appropriation of tens of millions of human artworks
7. **Humanism** GenAI is driven by Big Tech companies whose leaders are pushing an anti-humanist and anti-democratic agenda
8. **Reliability** An AI system always provides answers—including false or fabricated ones—and responds with problematic biases that replicate and amplify already existing ones (sexism, racism, etc.)
9. **Data centers** GenAI requires the construction of data centers, which have environmental impacts on nearby communities—often already disadvantaged—and lead to conflicts over the use of water and electricity
10. **Global Warming** GenAI contributes to the acceleration of global warming and destruction of ecosystems both directly and indirectly (through economic growth)

QUICK POLL

Fill the **Survey** for next week

- Python | R
- Notebooks (Rstudio, Jupyter, Org-Mode)
- Git
- Zenodo, Software Heritage
- Docker | Singularity
- Confidence Interval
- P-value, P-hacking
- Linear regression
- Model Selection
- Design of Experiments
- Reinforcement|Active Learning

TENTATIVE SCHEDULE

12 lectures Many practical homeworks (50%)

A final Exam (50%)

1. Presentation of the lecture
2. Conducting Experiments
3. Good Graphics
4. Conducting Experiments debrief
5. Analysis tools: probabilities, CI, estimation
 - Fall vacations
 - No lecture
6. Correlation/causality, the Linear model
7. The Linear model
 - No lecture
8. The academic publication system
9. Design of Experiments
10. Ethics and Scientific Integrity 101
 - Winter vacations
11. The academic publication system
12. Ethics (AI and humain, climate change, societal challenges)
 - Exam

Beautiful Viz with ggplot

Data management tools (tidyverse, git-annex)

TOWARDS A RIGOROUS AND ETHICAL COMPUTER SCIENCE

(Computational) Reproducible Research

- Journal and computational notebooks
- version control
- Containers

MOOC

Observation vs. Experiments

- Measurement / Tracing
- Modeling (parameters)
- Planning and Design of Experiments

Statistics

- Confidence interval and p-value
- Linear model and model selection
- Frequentist vs. Bayesian
- Test and inference

"Meta" Science

- Epistemology
- Scientific Integrity (fraud & errors)
- Publication, evaluation, and credit
- Ethics

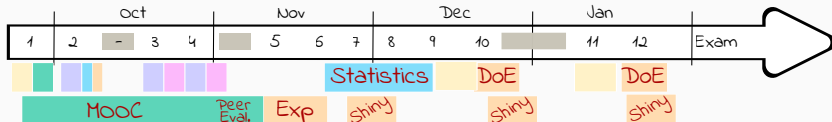
Data Management

- DMP, license
- File organisation, metadata
- version, archiving

Data Analysis/Visualization/Presentation

- R (python)
- Tidyverse (dplyr, ggplot)

MOOC



REPRODUCIBLE RESEARCH

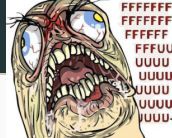
SCIENTIFIC CONSENSUS



NO TRANSPARENCY NO CONSENSUS



COMMON HORROR STORIES 1/3: *WHAT DID I DO?*



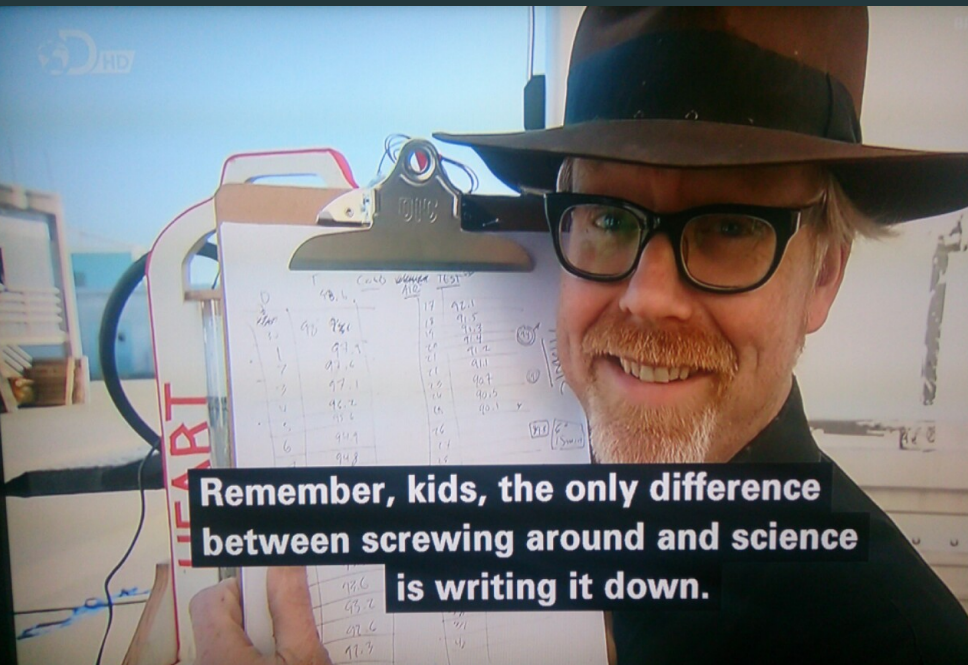
Author

- I thought I used the same parameters but I'm getting different results!
- The new student wants to compare with the method I proposed last year
- My advisor asked me whether I took care of setting this or this but I can't remember
- The damned fourth reviewer asked for a major revision and wants me to change Figure 3. Which code and which data set did I use?
- It worked yesterday! 6 months later: Why did I do that?

Reviewer

- As usual, there is no confidence interval, I wonder about the variability and whether the difference is significant or not
- That can't be true, I'm sure they removed some points
- Why is this graph in logscale? How would it look like otherwise? I'm not even sure of what this value means. If only I could access the generation script

MYTHBUSTERS: SCIENCE VS. SCREWING AROUND



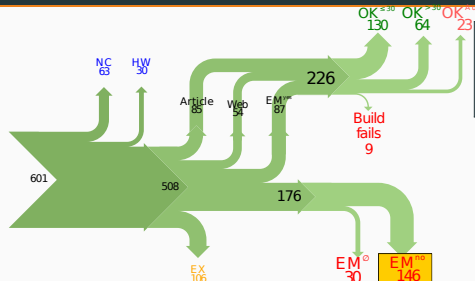
**Remember, kids, the only difference
between screwing around and science
is writing it down.**

COMMON HORROR STORIES 2/3: ARGH... DAMNED COMPUTERS

- **Alice:** I got 3.123123 **Bob:** I got segfault
- Damned! It used to work!!! Whenever I upgrade my computer, things break so I try to stay away from this 😞
- Whenever trying the code of my colleague, I had to install Foo but I broke everything and now neither his code nor mine works! 😞
- But hey! Here is my code. It's on GitHub so feel free to play with it! I'm doing open science 😊

Seriously ? It's 21st century. How come all this is so painful ?

COMMON HORROR STORIES 3/3: PLEASE HOLD ON



- Versioning Problems

Collberg, Christian et Al., *Measuring Reproducibility in Computer Systems Research*, <http://reproducibility.cs.arizona.edu/>

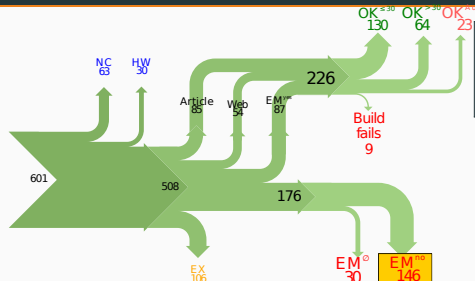
- 8 ACM conferences (ASPLOS'12, CCS'12, OOPSLA'12, OSDI'12, PLDI'12, SIGMOD'12, SOSP'11, VLDB'12) and 5 journals
- EM^{no} = the code cannot be provided

Thanks for your interest in the implementation of our paper. The good news is that I was able to find some code. I am just *hoping* that *it* is a stable working version of the code, and *matches the implementation we finally used for the paper*. Unfortunately, I have *lost some data* when *my laptop was stolen* last year. The bad news is that the code is not commented and/or clean.

10/14

Attached is the (system) source code of our algorithm. I'm *not very sure whether*

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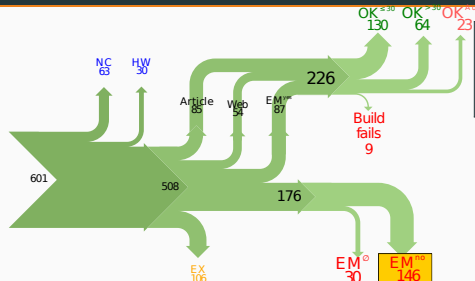


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- Versioning Problems
 - Bad Backup Practices

Unfortunately, the server in which my implementation was stored had a *disk crash in April and three disks crashed simultaneously*. While the help desk made significant effort to save the data, my entire implementation for this paper was not found.

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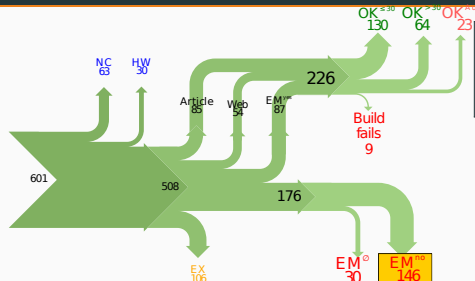


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 - Code Will be Available Soon

Unfortunately the current system is *not mature enough at the moment*, so it's not yet publicly available. We are actively working on a number of extensions and *things are somewhat volatile*. However, once things stabilize we plan to release it to outside users. At that point, we would be happy to send you a copy.

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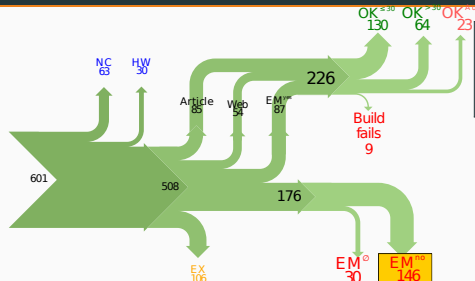


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- No Intention to Release

*I am afraid that the source code was never released. The code was **never intended to be released** so is not in any shape for general use.*

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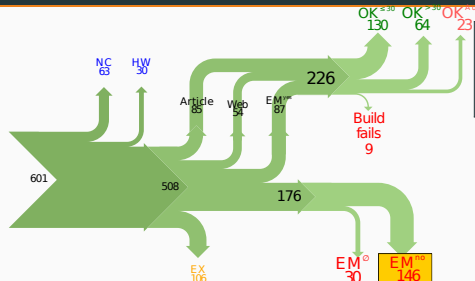
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- Programmer Left
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*<STUDENT> was a graduate student in our program but **he left a while back** so I am responding instead. For the paper we used a prototype that included many moving pieces that only <STUDENT> knew how to operate and we did not have the time to integrate them in a ready-to-share implementation before he left. Still, I hope you can build on the ideas/technique of the paper.*

Unfortunately, the author who has done most of the coding for this paper has

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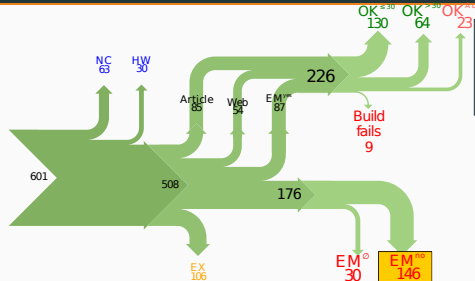
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- EM^{no} = the code cannot be provided
 - Programmer Left
 - Commercial Code
- Versioning Problems
- Bad Backup Practices
- Code Will be Available Soon
- No Intention to Release

Since this work has been done at $\langle \text{COMPANY} \rangle$ *we don't open-source code* unless there is a compelling business reason to do so. So unfortunately I don't think we'll be able to share it with you.

The code *owned by* $\langle \text{COMPANY} \rangle$, and AFAIK the code is not open-source. Your best bet is to reimplement :(Sorry.

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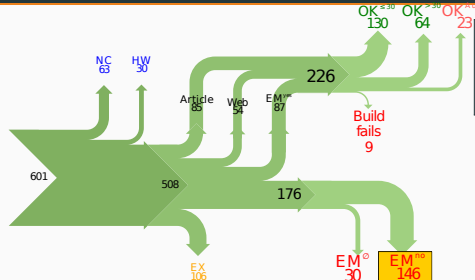
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- EM^{no} = the code cannot be provided
- Programmer Left
- Commercial Code
- Proprietary Academic Code

Unfortunately, the $\langle \text{SYSTEM} \rangle$ sources are *not meant to be opensource* (the code is partially *property of* $\langle \text{UNIVERSITY 1} \rangle$, $\langle \text{UNIVERSITY 2} \rangle$ and $\langle \text{UNIVERSITY 3} \rangle$.)

If this will change I will let you know, albeit I do not think there is an intention to make the $\langle \text{SYSTEM} \rangle$ sources opensource in the near future.

If you're interested in obtaining the code, *we only ask for a description of the re-*

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- EM^{no} = the code cannot be provided
 - Programmer Left
 - Commercial Code
 - Proprietary Academic Code
 - Research vs. Sharing
- Versioning Problems
- Bad Backup Practices
- Code Will be Available Soon
- No Intention to Release

In the past when we attempted to share it, we found ourselves spending more time getting outsiders up to speed than on our own research. So I finally had to establish the policy that we will not provide the source code outside the group.

DIFFERENT REPRODUCIBILITY CONCERNS IN MODERN SCIENCE

Social Sciences, Oncology, ... methodology, statistics, pre-registration

Genomics software engineering, computational reproducibility, provenance

Computational fluid dynamics numerical issues

Artificial Intelligence most of the above

The processing steps between raw observations and findings have gotten increasingly numerous and complex

Authors



Data

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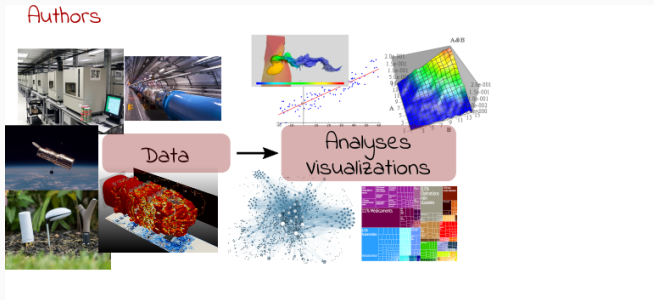
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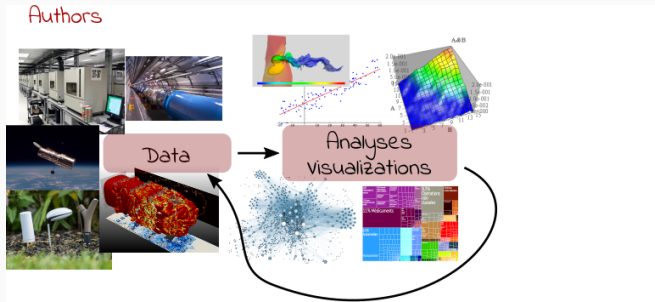
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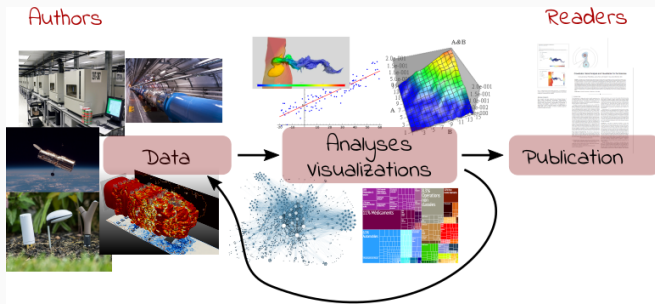
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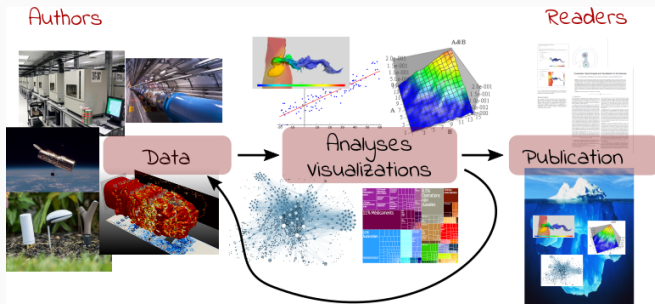
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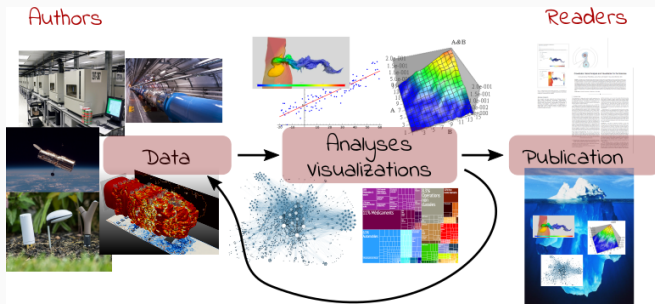
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Soft. Engineering, Statistics, and Reproducible Research in the **curricula**

Manifesto: *"I solemnly pledge"* (WSSSPE, Lorena Barba, FAIR)

1. I will teach my graduate students about reproducibility
2. All our research code (and writing) is under version control
3. We will always carry out verification and validation
4. We will share data, plotting script & figure under CC-BY
5. We will upload the preprint to arXiv at the time of submission of a paper
6. We will release code at the time of submission of a paper
7. We will add a "Reproducibility" declaration at the end of each paper
8. I will keep an up-to-date web presence



Learn and Teach using online resources like

- **Software Carpentry**, **The Turing Way**, ...

THE REPRODUCIBLE RESEARCH MOOC

MOOC Reproducible Research: Methodological principles for a transparent science, Learning Lab

- Konrad Hinsen, Christophe Pouzat
- 3rd Edition: March 2020 – ... (25,000+)
- In French, *fully subtitled in English*

Module 1 Taking notes (markdown, git)

Module 2 Computational document (Jupyter, Rstudio, Org-Mode)
Data Analysis of the Space Shuttle: Pre-Challenger Prediction of Failure

Module 3 Reproducible analysis
Data analysis : 7 possible subjects and a Peer evaluation

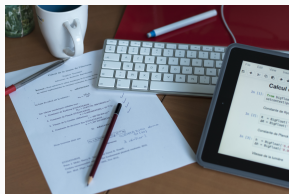
Module 4 Reproducibility Pitfalls (Hell)

The MOOC RR2: Practices and tools for managing computations and data

Module Managing data

Module Software environment control

Module Scientific workflow



HOMEWORKS

- ☐ Indicate your name on the Pad.
- ☐ Register on the Mattermost.
- ☐ Set up a **public** github or gitlab project for this lecture.
- ☐ Register to the MOOC RR1
- ☐ Follow modules 1 + 2 of the MOOC with as much exercises as possible
- ☐ Set up a computational document system (e.g., Rstudio or Jupyter)
- ☐ Report the URL of your git project, your mattermost ID on the Pad.
- ☐ Start learning R by reading my short *R crash course for computer scientists*